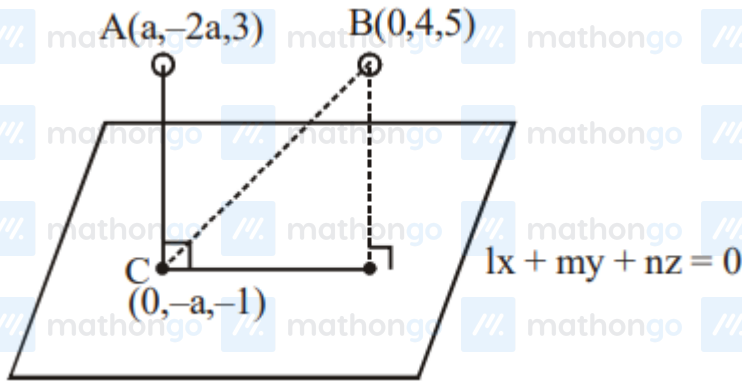


Q1 (4)



C lies on plane $\Rightarrow -ma - n = 0 \Rightarrow \frac{m}{n} = -\frac{1}{a}, \dots, (1)$

$\vec{CA} \parallel \hat{i} + m\hat{j} + n\hat{k}$

$$\frac{a-0}{1} = \frac{-a}{m} = \frac{4}{n} \Rightarrow \frac{m}{n} = -\frac{a}{4}$$

From (1) & (2)

$$-\frac{1}{a} = \frac{-a}{4} \Rightarrow a^2 = 4 \Rightarrow a = 2 \quad (\text{since } a > 0)$$

$$\text{From (2)} \quad \frac{m}{n} = \frac{-1}{2}$$

$$\text{Let } m = -t \Rightarrow n = 2t$$

$$\frac{2}{1} = \frac{-2}{-t} \Rightarrow 1 = t$$

$$\text{So plane : } t(x - y + 2z) = 0$$

$$BD = \frac{6}{\sqrt{6}} = \sqrt{6} \quad C \cong (0, -2, -1)$$

$$CD = \sqrt{BC^2 - BD^2}$$

$$= \sqrt{(0^2 + 6^2 + 6^2) - (\sqrt{6})^2}$$

$$= \sqrt{66}$$

Q2 (2)

$$P(3, -1, 2)$$

$$Q(1, 2, -4)$$

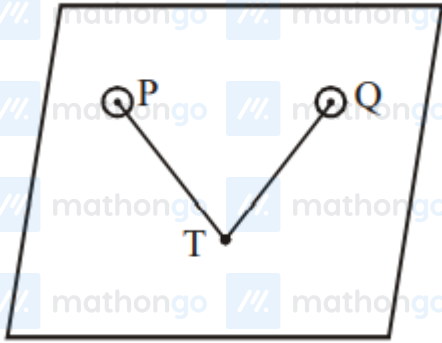
$$\vec{PR} \parallel 4\hat{i} - \hat{j} + 2\hat{k}$$

$$\vec{QS} \parallel -2\hat{i} + \hat{j} - 2\hat{k}$$

dr's of normal to the plane containing

P, T & Q will be proportional to

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & -1 & 2 \\ -2 & 1 & -2 \end{vmatrix}$$



$$\therefore \frac{\ell}{0} = \frac{m}{4} = \frac{n}{2}$$

$$\text{For point, T : } \vec{PT} = \frac{x-3}{4} = \frac{y+1}{-1} = \frac{z-2}{2} = \lambda$$

$$\vec{QT} = \frac{x-1}{-2} = \frac{y-1}{1} = \frac{z+4}{-2} = \mu$$

$$T : (4\lambda + 3, -\lambda - 1, 2\lambda + 2)$$

$$\cong (2\mu + 1, \mu + 2, -2\mu - 4)$$

$$4\lambda + 3 = -2\mu + 1 \Rightarrow 2\lambda + \mu = -1$$

$$\lambda + \mu = -3 \Rightarrow \lambda = 2$$

$$\& \mu = -5 \quad \lambda + \mu = -3 \Rightarrow \lambda = 2$$

$$\text{So point T : } (11, -3, 6)$$

$$\vec{OA} = (11\hat{i} - 3\hat{j} + 6\hat{k}) \pm \left(\frac{2\hat{j} + \hat{k}}{\sqrt{5}}\right)\sqrt{5}$$

$$\vec{OA} = (11\hat{i} - 3\hat{j} + 6\hat{k}) \pm (2\hat{j} + \hat{k})$$

$$\vec{OA} = 11\hat{i} - \hat{j} + 7\hat{k}$$

or

$$9\hat{i} - 5\hat{j} + 5\hat{k}$$

$$|\vec{OA}| = \sqrt{121 + 1 + 49} = \sqrt{171}$$

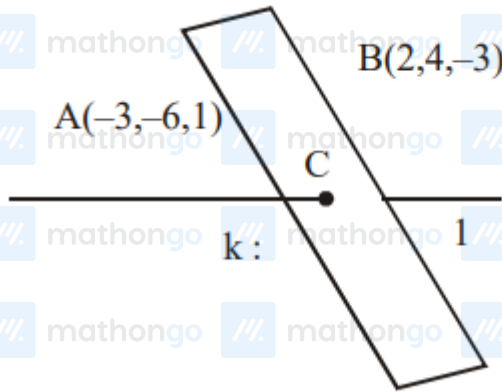
Hints and Solutions

MathonGo

or

$$\sqrt{81 + 25 + 25} = \sqrt{131}$$

Q3 (3)



Point C is

$$\left(\frac{2k-3}{k+1}, \frac{4k-6}{k+1}, \frac{-3k+1}{k+1}\right)$$

$$\frac{x-1}{-1} = \frac{y+4}{2} = \frac{z+2}{3}$$

Plane $lx + my + nz = 0$

$$l(-1) + m(2) + n(3) = 0$$

$$-1 + 2m + 3n = 0$$

It also satisfy point $(1, -4, -2)$ $l - 4m - 2n = 0$ Solving (1) and (2) $2m + 3n = 4m + 2n$

$$n = 2m$$

$$l - 4m - 4m = 0$$

$$l = 8m$$

$$\frac{l}{8} = \frac{m}{1} = \frac{n}{2}$$

$$l : m : n = 8 : 1 : 2$$

Plane is $8x + y + 2z = 0$ It will satisfy point C $8\left(\frac{2k-3}{k+1}\right) + \left(\frac{4k-6}{k+1}\right) + 2\left(\frac{-3k+1}{k+1}\right) = 0$

$$16k - 24 + 4k - 6 - 6k + 2 = 0$$

$$14k = 28 \quad \therefore k = 2$$

Q4 (2)

Plane passing through $(42, 0, 0), (0, 42, 0),$
 $(0, 0, 42)$

From intercept form, equation of plane is

$$x + y + z = 42$$

$$\Rightarrow (x - 11) + (y - 19) + (z - 12) = 0$$

let $a = x - 11, b = y - 19, c = z - 12$

$$a + b + c = 0$$

Now, given expression is

$$3 + \frac{a}{b^2c^2} + \frac{b}{a^2c^2} + \frac{c}{a^2b^2} - \frac{42}{14abc}$$

$$3 + \frac{a^3 + b^3 + c^3 - 3abc}{a^2b^2c^2}$$

If $a + b + c = 0$

$$\Rightarrow a^3 + b^3 + c^3 = 3abc$$

$$\Rightarrow 3$$

Q5 (2)

$(3, 5, 7)$ satisfy the line L_1

$$\frac{3-a}{\ell} = \frac{5-2}{3} = \frac{7-b}{4}$$

$$\frac{3-a}{\ell} = 1 \quad \&$$

$$\frac{7-b}{4} = 1$$

$$a + \ell = 3 \quad \dots (1)$$

$$\& \quad b = 3 \dots (2)$$

$$\vec{v}_1 = \langle 4, 3, 8 \rangle \quad \& \quad \langle 3, 5, 7 \rangle$$

$$\vec{v}_1 = \langle 1, -2, 1 \rangle$$

$$\vec{v}_2 = \langle \ell, 3, 4 \rangle$$

$$\vec{v}_1 \cdot \vec{v}_2 = 0 \Rightarrow \ell - 6 + 4 = 0 \Rightarrow \ell = 2$$

$$a + \ell = 3 \Rightarrow a = 1$$

Hints and Solutions

MathonGo

$$L_1 : \frac{x+1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$$

$$L_2 : \frac{x-2}{3} = \frac{y-4}{4} = \frac{z-5}{5}$$

$$A = \langle 1, 2, 3 \rangle$$

$$B = \langle 2, 4, 5 \rangle$$

$$\vec{AB} = \langle 1, 2, 2 \rangle$$

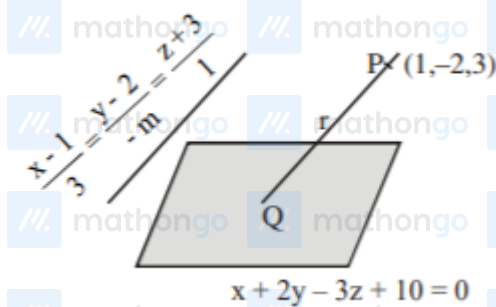
$$\vec{p} = 2\hat{i} + 3\hat{j} + 4\hat{k}$$

$$\vec{q} = 3\hat{i} + 4\hat{j} + 5\hat{k}$$

$$\vec{p} \times \vec{q} = -\hat{i} + 2\hat{j} - \hat{k}$$

$$\text{Shortest distance} = \left| \frac{\vec{AB} \cdot (\vec{p} \times \vec{q})}{|\vec{p} \times \vec{q}|} \right| = \frac{1}{\sqrt{6}}$$

Q6 (2)



$$\text{DC of line} \equiv \left(\frac{3}{\sqrt{m^2+10}}, \frac{-m}{\sqrt{m^2+10}}, \frac{1}{\sqrt{m^2+10}} \right)$$

$$Q \text{ lies on } x + 2y - 3z + 10 = 0$$

$$1 + \frac{3r}{\sqrt{m^2+10}} - 4 - \frac{2mr}{\sqrt{m^2+10}} - 9 - \frac{3r}{\sqrt{m^2+10}} + 10 = 0$$

$$\Rightarrow \frac{r}{\sqrt{m^2+10}} (3 - 2m - 3) = 2$$

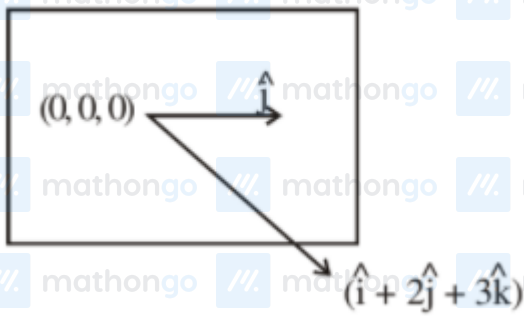
$$\Rightarrow \frac{r}{\sqrt{m^2+10}} (-2m) = 2$$

$$r^2 m^2 = m^2 + 10$$

$$\frac{7}{2} m^2 = m^2 + 10 \Rightarrow \frac{5}{2} m^2 = 10 \Rightarrow m^2 = 4$$

$$|m| = 2$$

Q7 (4)



$$\vec{n} = \hat{j} \times (\hat{i} + 2\hat{j} + 3\hat{k})$$

$$= -3\hat{i} + 0\hat{j} + \hat{k}$$

$$\text{So, } (-3)(x - 1) + 0(y - 2) + (1)(z - 3) = 0$$

$$\Rightarrow -3x + z = 0$$

Option 4

Alternate :

Required plane is

$$\begin{vmatrix} x & y & z \\ 0 & 1 & 0 \\ 1 & 2 & 3 \end{vmatrix} = 0$$

$$\Rightarrow 3x - z = 0$$

Q8 (4)

$$\text{Required plane is } p_1 + \lambda p_2 = (2 + 3\lambda)x - (7 + 5\lambda)y$$

$$+ (4 + 4\lambda)z - 3 + 11\lambda = 0$$

which is satisfied by $(-2, 1, 3)$.

$$\text{Hence, } \lambda = \frac{1}{6}$$

$$\text{Thus, plane is } 15x - 47y + 28z - 7 = 0$$

$$\text{So, } 2a + b + c - 7 = 4$$

Q9 (2)

$$\therefore \text{Reflection } (-2, 4, -6) \text{ Plane : } \begin{vmatrix} x-2 & y-1 & z+1 \\ 3 & -2 & 1 \\ 4 & -3 & 5 \end{vmatrix} = 0$$

$$\Rightarrow (x-2)(-10+3) - (y-1)(15-4) + (z+1)(-1) = 0$$

$$\Rightarrow -7x + 14 - 11y + 11 - z - 1 = 0$$

$$\Rightarrow 7x + 11y + z = 24$$

$$\therefore \alpha = 7, \beta = 11, \gamma = 1$$

$$\alpha + \beta + \gamma = 19$$

Q10 (0)

Let point P is (α, β, γ)

$$\left(\frac{\alpha + \beta + \gamma}{\sqrt{3}}\right)^2 + \left(\frac{\ell\alpha - n\gamma}{\sqrt{\ell^2 + n^2}}\right)^2 + \left(\frac{\alpha - 2\beta + \gamma}{\sqrt{6}}\right)^2 = 9$$

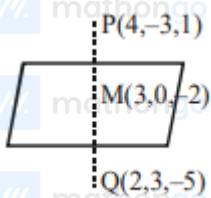
Locus is

$$\frac{(x+y+z)^2}{3} + \frac{(\ell x - n z)^2}{\ell^2 + n^2} + \frac{(x-2y+z)^2}{6} = 9$$

$$x^2 \left(\frac{1}{2} + \frac{\ell^2}{\ell^2 + n^2}\right) + y^2 + z^2 \left(\frac{1}{2} + \frac{n^2}{\ell^2 + n^2}\right) + 2zx \left(\frac{1}{2} - \frac{\ell n}{\ell^2 + n^2}\right) - 9 = 0$$

Since its given that $x^2 + y^2 + z^2 = 9$ After solving $\ell = n$

Q11 (28)



$$\text{Plane is } 1(x-3) - 3(y-0) + 3(z+2) = 0$$

Hints and Solutions

MathonGo

$$x - 3y + 3z + 3 = 0$$

$$(a^2 + b^2 + c^2 + d^2)_{\min} = 28$$

Q12 (4)

Let plane is $x - 2y + 2z + \lambda = 0$

distance from $(1, 2, 3) = 1$

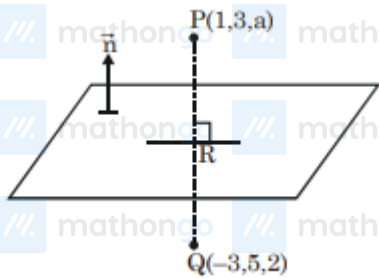
$$\Rightarrow \frac{|\lambda + 3|}{5} = 1 \Rightarrow \lambda = 0, -6$$

$$\Rightarrow a = 1, b = -2, c = 2, d = -6 \text{ or } 0$$

$$b - d = 4 \text{ or } -2, c - a = 1$$

$$\Rightarrow k = 4 \text{ or } -2$$

Q13 (1)



$$\text{plane} = 2x - y + z = b$$

$$R \equiv (-1, 4, \frac{a+2}{2}) \rightarrow \text{on plane}$$

$$\therefore -2 - 4 + \frac{a+2}{2} = b$$

$$\Rightarrow a + 2 = 2b + 12 \Rightarrow a = 2b + 10 \dots (i)$$

\$

$$= \langle 4, -2, a-2 \rangle \therefore \frac{2}{4} = \frac{-1}{-2} = \frac{1}{a-2} \rightarrow a-2=2 \rightarrow a=4, b=-3$$

$$\therefore |a+b| = 1$$

Q1: 16 March (Shift 1) - Single Correct

If for $a > 0$, the feet of perpendiculars from the points $A(a, -2a, 3)$ and $B(0, 4, 5)$ on the plane $lx + my + nz = 0$ are points $C(0, -a, -1)$ and D respectively, then the length of line segment CD is equal to :

(1) $\sqrt{31}$

(2) $\sqrt{41}$

(3) $\sqrt{55}$

(4) $\sqrt{66}$

Q2: 16 March (Shift 1) - Single Correct

Let the position vectors of two points P and Q

be $3\hat{i} - \hat{j} + 2\hat{k}$ and $\hat{i} + 2\hat{j} - 4\hat{k}$, respectively. Let R and S be two points such that the direction ratios of lines PR and QS are $(4, -1, 2)$ and $(-2, 1, -2)$, respectively. Let lines PR and QS intersect at T . If the vector $\vec{TA}$ is perpendicular to both $\vec{PR}$ and $\vec{QS}$ and the length of vector $\vec{TA}$ is $\sqrt{5}$ units, then the

modulus of a position vector of A is :

(1) $\sqrt{482}$

(2) $\sqrt{171}$

(3) $\sqrt{5}$

(4) $\sqrt{227}$

Q3: 16 March (Shift 1) - Single Correct

Let P be a plane $lx + my + nz = 0$ containing the line, $\frac{1-x}{1} = \frac{y+4}{2} = \frac{z+2}{3}$. If plane P divides the line segment AB joining points $A(-3, -6, 1)$ and $B(2, 4, -3)$ in ratio $k : 1$ then the value of k is equal to :

(1) 1.5

(2) 3

(3) 2

(4) 4

Q4: 16 March (Shift 2) - Single Correct

If (x, y, z) be an arbitrary point lying on a plane P which passes through the point $(42, 0, 0)$, $(0, 42, 0)$ and $(0, 0, 42)$, then the value of expression

$$3 + \frac{x-11}{(y-19)^2(z-12)^2} + \frac{y-19}{(x-11)^2(z-12)^2} + \frac{z-12}{(x-11)^2(y-19)^2} - \frac{x+y+z}{14(x-11)(y-19)(z-12)}$$

(1) 0

(2) 3

(3) 39

(4) -45

Q5: 16 March (Shift 2) - Single Correct

If the foot of the perpendicular from point $(4, 3, 8)$ on the line $L_1: \frac{x-a}{l} = \frac{y-2}{3} = \frac{z-b}{4}$, $l \neq 0$ is $(3, 5, 7)$, then the shortest distance between the line L_1 and line $L_2: \frac{x-2}{3} = \frac{y-4}{4} = \frac{z-5}{5}$ is equal to :

(1) $\frac{1}{2}$ (2) $\frac{1}{\sqrt{6}}$ (3) $\sqrt{\frac{2}{3}}$ (4) $\frac{1}{\sqrt{3}}$

Q6: 16 March (Shift 2) - Numerical

If the distance of the point $(1, -2, 3)$ from the plane $x + 2y - 3z + 10 = 0$ measured parallel to the line,

$$\frac{x-1}{3} = \frac{2-y}{m} = \frac{z+3}{1} \text{ is } \sqrt{\frac{7}{2}}, \text{ then the value of } |m| \text{ is equal to } \underline{\hspace{2cm}}$$

Q7: 17 March (Shift 1) - Single Correct

The equation of the plane which contains the y-axis and passes through the point $(1, 2, 3)$ is :

Questions with Answer Keys

MathonGo

(1) $x + 3z = 10$

(2) $x + 3z = 0$

(3) $3x + z = 6$

(4) $3x - z = 0$

Q8: 17 March (Shift 1) - Numerical

If the equation of the plane passing through the line of intersection of the planes

$$2x - 7y + 4z - 3 = 0, 3x - 5y + 4z + 11 = 0$$

and the point $(-2, 1, 3)$ is $ax + by + cz - 7 = 0$, then the value of $2a + b + c - 7$ is ____.

Q9: 17 March (Shift 2) - Single Correct

If the equation of plane passing through the

mirror image of a point $(2, 3, 1)$ with respect to line $\frac{x+1}{2} = \frac{y-3}{1} = \frac{z+2}{-1}$ and containing the line

$$\frac{x-2}{3} = \frac{1-y}{2} = \frac{z+1}{1} \text{ is } \alpha x + \beta y + \gamma z = 24 \text{ then } \alpha + \beta + \gamma \text{ is equal to :}$$

(1) 20

(2) 19

(3) 18

(4) 21

Q10: 17 March (Shift 2) - Numerical

Let P be an arbitrary point having sum of the squares of the distance from the planes

$x + y + z = 0$, $lx - nz = 0$ and $x - 2y + z = 0$ equal to 9. If the locus of the point P is $x^2 + y^2 + z^2 = 9$, then the value of $l - n$ is equal to ____.

Q11: 18 March (Shift 1) - Numerical

Let the plane $ax + by + cz + d = 0$ bisect the line joining the points $(4, -3, 1)$ and $(2, 3, -5)$ at the right angles. If

a, b, c, d are integers, then the minimum value of $(a^2 + b^2 + c^2 + d^2)$ is

Q12: 18 March (Shift 1) - Numerical

The equation of the planes parallel to the plane $x - 2y + 2z - 3 = 0$ which are at unit distance from the point $(1, 2, 3)$ is $ax + by + cz + d = 0$. If $(b - d) = K(c - a)$, then the positive value of K is

Q13: 18 March (Shift 2) - Numerical

Let the mirror image of the point $(1, 3, a)$ with respect to the plane $\vec{r} \cdot (2\hat{i} - \hat{j} + \hat{k}) - b = 0$ be $(-3, 5, 2)$. Then the value of $|a + b|$ is equal to ____

Answer Key**Q1 (4)****Q2 (2)****Q3 (3)****Q4 (2)****Q5 (2)****Q6 (2)****Q7 (4)****Q8 (4)****Q9 (2)****Q10 (0)****Q11 (28)****Q12 (4)****Q13 (1)**